



- 2 (a) State what is meant by the spring constant of a spring.

.....
..... [1]

- (b) A block of mass 0.086 kg is suspended in equilibrium from a spring that obeys Hooke's law. In this position, the extension of the spring is 0.026 m.

The block is pulled vertically down a distance of 0.015 m from the equilibrium position and then released from rest so that it starts moving upwards. Air resistance is negligible.

- (i) Identify **one** possible source of random error in determining experimentally the extension of the spring using a ruler.

.....
..... [1]

- (ii) Calculate the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

- (iii) Show that, in the equilibrium position, the elastic potential energy of the spring is 0.011 J.

[2]

- (iv) Assume that the spring has not exceeded its elastic limit when the block is in the position from which it is released.

Calculate the additional elastic potential energy stored in the spring when the block moves from its equilibrium position to the position from which it is released.

additional elastic potential energy = J [2]

- (v) Use the principle of conservation of energy to determine the kinetic energy of the block as it passes the equilibrium position moving upwards.

kinetic energy = J [2]

- (c) Suggest, with a reason, how the answer in **2(b)(iv)** is affected if the spring in **2(b)** has exceeded its elastic limit when the block is at the position from which it is released.

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..... [2]

[Total: 12]